Footprint borders facilitator guide

Mathematical overview

Object. A border is the ideal infinite repetition in both directions of a finite motif arrangement, bounded across the strip. Only footprints and their orientations count. Ends, guide lines and temporary labels do not. The intended examples are discrete, nonempty patterns with a shortest positive translation P along the border. Finite scraps alone do not establish an infinite symmetry.

Glide theorem. Write $H(x,y) = (x,-y)$ for reflection in the middle line and $T_a(x,y) = (x+a,y)$ for a slide along it. They commute. A glide $G_a = T_a H$ with $a \neq 0$ satisfies $G_a^2 = T_{2a}$. Thus a matching 3 cm glide forces a matching 6 cm slide, but need not force a 3 cm slide or the bare reflection H.

Shortest distances. If a border has a glide along this line but H itself fails, its shortest glide length is P/2. Every glide displacement is $(k + 1/2)P$, k an integer. If H is a symmetry instead, matching glide displacements are kP; the shortest nonzero glide length is P. These statements require a primitive translation P and a fixed glide axis; an arbitrary matching glide need not be the shortest.

Forced motions. Reflections in two perpendicular lines compose to a half-turn about their crossing. The printed crossing lines are perpendicular. For general crossing lines the rotation angle is twice their angle; a half-turn is not a theorem for arbitrary crossing lines. Every composition or inverse of symmetries is a symmetry.

Limits. There are seven discrete frieze symmetry types, but these problems do not prove or require that classification. A simple motif can accidentally add symmetries. The specified notched F-like motif has no nonidentity planar isometry, so its handedness and direction distinguish candidate moves.

What children are meant to discover

- **K-1:** Build repeating borders and use a copy to distinguish slides, flips, glides and half-turns. A local matching attempt is evidence to investigate the repeated rule.
- **Grades 2-3:** Find a glide that works although the flip alone fails; compare shortest distances and change which motions a border permits.
- **Grades 4-5:** Explain forced symmetries by composition and justify a full symmetry description for one chosen construction. Sampling several moves is not a proof of completeness.

Status. The current K-1 packet has seven problems on five pages; the two older v3 packets retain six problems each. Mathematical examples are verified; physical cutout handling and classroom pacing are untested. The Grades 2-3 edit explicitly permits moved footprints and a larger repeating part.

Preparation and session plan

Materials and readiness

For eleven children, use three adult-led tables. Three-child groups can rotate builder, tester and observer; budget five pair-sized material sets overall. Per pair supply 20-24 reusable footprint pieces, four paper strips at least 70 by 280 mm, two tracing strips, a pencil or washable marker, and a ruler. Provide a 600 mm joined demonstration strip per table. Keep repeated motifs separated and use at least three or four visible repeats.

Motifs: use the exact notched F-like outline printed in the introduction, at least 25 mm long, with its direction arrow. Supply the original, its horizontal mirror, its vertical mirror and its half-turn; ordinary symmetric footprints can change the answers. An economical alternative is identical rectangular cards carrying those drawings. State that card edges and blank backgrounds are handling aids, not pattern data. Prepare visible marks on both sides or use tracing paper so a reflection can be enacted accurately. Adults cut and prepare pieces, not children during the investigation.

Preparation estimate: 30-45 minutes for a first reusable set, substantially less when reusing it. Before class, check that the chosen pieces really distinguish all four forms, test one reflected tracing strip, and rehearse the common launch. The packet is not a complete manipulative kit. If the exact outline is hard to reproduce, trace and copy its printed example rather than substitute a symmetric stamp without checking it.

Readiness: children need to extend a repeating arrangement and compare a moved copy with every motif in the original. Reading and measured lengths can be adult-supported; the youngest can compare slide lengths directly. If keeping the original fixed or mentally continuing the ends is difficult, spend the session on one border and slide matching.

Common launch

Let children handle pieces first. Start K-1 with one repeating row and a tracing copy: slide the copy to a matching position and a position that fails. Keep the original fixed. Introduce the single-motif flip-then-slide visual only before the first glide task. Keep the label A with the same piece during the demonstration. This teaches the move without giving a target border. Compare two or three interior repetitions; explain that the printed ends are a window on an endless pattern. Demonstrate a half-turn about a chosen point when first needed, including a point between motifs.

Flexible hour

0-8 minutes: handling and launch. **8-23:** two different shortest periods, then the glide challenge. Let a successful pattern remain on one strip while a second is built. **23-27:** movement break, using a hand-held copy to show a slide and a half-turn. **27-45:** selected reflection/half-turn construction or edit. **45-56:** partner testing and one explanation of a forced motion. **56-60:** save the generating rule and note what children could actually justify.

A good stopping point is one border with a well-defended unusual symmetry. Upper Problem 6 can occupy an entire later discussion. Offer hints only when needed: “Where must this one footprint land?” then “Does its direction agree?” then “What happens if you do your matching move twice?”

Concrete answer bank

A precise way to read these examples

Let the middle line be y = 0. Let **A** be the printed asymmetric footprint, **H A** its reflection across a horizontal line, **V A** its reflection across a vertical line, and **R A** its half-turn. These names describe the entire footprint, including its direction arrow. Positions below are motif anchors in centimeters; translate the same anchor consistently. Take h = 1.5 cm and use the original motif at about 26 mm wide. In every recipe, k is any integer, so the pattern continues both ways.

Border T, slides only: put A at (6k, h). The primitive period is 6 cm. A variant with A at (4k, h) has primitive period 4 cm. A translation must take an A anchor to another A anchor, so no smaller positive slide works. A half-turn would reverse the asymmetric motif; no reversed motif is present.

Border G, a glide without its bare reflection: put A at (6k, h) and H A at (6k + 3, -h). A 3 cm rightward glide takes an upper A to the next lower H A and that lower piece to the next upper A. The shortest glide is 3 cm and the shortest slide is 6 cm. A bare horizontal flip lands in the gaps. A 3 cm bare slide lands on the wrong row. No vertical reflection or half-turn matches because it would introduce V A or R A, neither of which occurs.

Border H, a horizontal reflection without any half-turn: put A at (6k, h) and H A at (6k, -h). The middle-line flip matches paired motifs. Every half-turn changes the direction/handedness to R A or V A, forms absent from the border. Horizontal glides by nonzero multiples of 6 cm also work.

Border R, half-turns without reflections: put A at (6k, h) and R A at (6k + 3, -h). Half-turn about (1.5, 0) matches, and so do half-turns about (1.5 + 3k, 0). The shortest slide is 6 cm. Horizontal or vertical reflection introduces H A or V A, so neither type of reflection matches. No glide works either.

Border E, both perpendicular reflections: repeat four motifs every 6 cm: A at (1.5, h), V A at (-1.5, h), H A at (1.5, -h), and R A at (-1.5, -h). Reflection in y = 0 and in x = 0 both match. Their composition is the half-turn about (0,0). The primitive slide is 6 cm; a 3 cm slide preserves locations but changes the required motif orientations.

Checking a child's own border

These are witnesses, not the only solutions. Ask for the repeating part and test its seams as well as its interior. A drawing with three locally similar motifs may fail at a join. A complete argument uses a generating rule and shows where every kind of motif lands. When excluding a motion, a single unavoidable mismatch is enough. Do not infer the shortest period from the length of the block the child happened to draw.

Keys for K-1 and Grades 2-3

K-1 continuation, Problems 2-7

Problem 2: Border T with 4 cm spacing and with 6 cm spacing gives two different shortest slides. The child need not name centimeters; compare the moved copies. Repeating a 4 cm block twice does not make an 8 cm primitive period.

Problem 3: Border G is a witness: a flip and 3 cm slide works; the flip alone fails. It is enough for a child to demonstrate those two facts on several repeats and describe the continuation.

Problem 4: Another G border with top motifs at 8k and bottom mirrored motifs at 8k + 4 has the same kind of match but a different size. Alternatively change the motif's vertical distance from the middle line while keeping the paired offsets consistent. Preserve the nonzero parallel slide requirement.

Problem 5: Start with Border H. Shift every lower-row mirrored footprint 3 cm right to obtain Border G; the old middle-line reflection now fails. Repeat the same edit in every 6 cm block so the new border still repeats.

Problem 6: Border R matches after a half-turn about (1.5,0). Border T has no matching half-turn about any point, because no half-turned footprint occurs anywhere. Testing just a printed tick as a center is not sufficient.

Problem 7: Answers depend on the traded borders. Use the recipes on page 3 as checks, not as a required catalog. Encourage the tester to show a matching move and a failed move, with the whole repeat accounted for.

Grades 2-3

Problem 1: Border T at 4 cm and 6 cm spacings. Mark the displacement of a particular motif on the tracing copy. Its image must be the nearest next A for the shortest positive match.

Problem 2: Border G. Neither the bare horizontal flip nor the 3 cm bare slide works, although their glide does.

Problem 3: On every border satisfying Problem 2, the shortest glide is half the shortest slide. In G these are 3 and 6 cm. This is a theorem under the stated discrete-period assumptions, not just a feature of the example; proof on page 6.

Problem 4: Border H has the requested flip and no half-turn about any point. The asymmetric motif rules out the reversed orientations a half-turn would need.

Problem 5: Border R works. Its middle-line flip fails. The task does not demand the absence of all other reflections, but this witness has none.

Problem 6: One simple edit is H to G above, breaking the old reflection and keeping period 6. To break an old translation instead, start with T at $x = 6k$. In every 12 cm block move the footprint at $x = 12k$ to $x = 12k + 0.5$, leaving the one at $x = 12k + 6$ unchanged. The new gaps alternate 5.5 and 6.5 cm. The old 6 cm slide fails and a 12 cm slide works. This edit uses the explicitly allowed larger repeating part; it is not a one-off defect in an otherwise endless unchanged border.

Key for Grades 4-5

Problem 1: Border T with periods 4 and 6 cm. Since each motif is asymmetric, a matching translation preserves its form and takes an anchor to an anchor. Consecutive anchors are precisely 4 or 6 cm apart, excluding smaller positive translations. An unnecessarily large chosen repeat is not evidence of a large primitive period.

Problem 2: Border G has shortest slide 6 cm and shortest glide 3 cm. Tracking one upper A forces slide displacement into 6 times an integer, or glide displacement into 3 plus 6 times an integer. Each permitted displacement also maps the lower mirrored row correctly.

Problem 3: A 6 cm slide must work: apply the given 3 cm glide twice. A 3 cm slide need not work; Border G is a counterexample. Do not infer that 6 is necessarily the shortest matching slide, since the hypothesis only gives one matching glide.

Problem 4: Border E is a construction. The perpendicular flips force a half-turn at their intersection. As the border already repeats, composing these flips and the period generates further symmetries: translated copies of the perpendicular mirror line, intermediate perpendicular mirror lines, half-turn centers, and glides formed from the middle-line flip and translations. For E, reflections occur in $y = 0$ and in $x = 3k$, half-turn centers at $(3k,0)$, slides by $6k$, and glides along $y = 0$ by nonzero $6k$. The two crossing flips alone do not produce a nonzero translation; the periodicity is a separate given feature.

Problem 5: Border R matches by a half-turn and has neither a middle-line reflection nor any perpendicular reflection. At an upper A, either type of reflection would require one of the missing mirror forms. The lack of those forms rules out every possible perpendicular axis, not only the axis through the displayed center.

Problem 6: Answers are construction-dependent. A complete answer for G is: translations by $6k$ cm ($k \neq 0$ for a moving slide), glides along $y = 0$ by $(6k + 3)$ cm, no pure reflection, and no nonidentity rotation. Include the identity only if discussing the full group. For R: translations by $6k$ cm and half-turns about $(1.5 + 3k,0)$, with no reflections or glides. Negative displacements are included.

Why these descriptions are complete

For these nonempty bounded-width discrete borders, the unbounded direction is the horizontal direction. An isometry preserving the border must preserve that direction, so its linear part is one of the four sign changes: identity, horizontal reflection, vertical reflection, or a half-turn. A nonzero vertical translation is impossible for a bounded-width pattern. The symmetric two-row examples force the line or centers to have $y = 0$. The asymmetric motif then forces the listed orientation, and its anchor forces exactly the stated x offsets. This is an adult completeness argument; children can explain the corresponding anchor-and-direction obstruction concretely.

Optional extension: Compare H and G. Both have matching glides, but only H has the bare reflection. Ask which shortest-distance relationship changes and why. Seven frieze names are not needed to settle this question.

Proofs and source notes

Shortest glide theorem

The translations form P times the integers. Suppose G_a is a symmetry along y = 0. Because H and horizontal translations commute, $G_a^2 = T_{2a}$, so 2a is an integer multiple of P. Composing G_a with any matching translation changes its displacement by a multiple of P. Reduce a modulo P into [0,P). The only possibilities are 0 and P/2. If the remainder were 0, composing with the inverse translation would make H itself a symmetry. Thus when H fails, the remainder is P/2 and the least absolute nonzero glide displacement is P/2. Any two glides along the same axis differ by a matching translation, so all their displacements are (k + 1/2)P.

If H is a symmetry, composing G_a with H gives T_a; hence glide displacements are precisely translation displacements. Conversely, every translation composed with H gives a glide when the displacement is nonzero. Thus the shortest nonzero glide length is P in that case.

What the finite checks establish

The independent guide check represents the exact notched polygon by integer-coordinate vertices and verifies that no nonidentity cyclic rotation or reversal of its vertices preserves all pairwise distances. It then checks each construction by exact positions and oriented motif type in a periodic cell, including the larger-repeat edit. This verifies these recipes and their primitive periods. The endless symmetry statements follow from their repetition rules and the geometric argument on page 5, not from a finite picture's endpoints.

Important teaching distinctions: a symmetry of a footprint is not the same as a symmetry of the arrangement; a matching glide need not split into two matching component moves; a long drawn repeat need not be primitive; and failure around one chosen center does not rule out every half-turn.

Sources

Joseph A. Gallian, **Contemporary Abstract Algebra**, author-hosted Chapter 28, **Frieze Groups and Crystallographic Groups**, pp. 446-449, especially pattern II, its squared-glide translation subgroup, and pattern VII's perpendicular reflections. Author-hosted chapter. This source supplies the symmetry-group context; the exact notched motif, constructions and shortest-period proof here are independently developed. The full seven-type classification is background only.

Laura Givental, Maria Nemirovskaya and Ilya Zakharevich, **Math Circle by the Bay**, AMS/MSRI Mathematical Circles Library 21 (2018), Preface pp. viii-ix. It describes substantial themes, manipulatives, dialogue and varied pace. The proposed one-hour sequence, exact motif preparation and readiness gates here are local plans, not a report of testing these tasks in that source.

Packet basis: current Week 35 Footprint borders: K-1 v2, Problems 1-7; Grades 2-3 and 4-5 v3, Problems 1-6. This guide uses the revised shortest-period wording, freely chosen half-turn centers, excluded guide lines and enlarged-repeat edit convention.

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

K-1 first route: Problem 1 makes one repeating border and tests matching and failing slides with a tracing copy; Problem 2 compares two shortest slides when the first action is reliable. Keep at least three or four visible repeats and the original fixed. Introduce the combined flip-and-slide only before Problems 3-4. On return, K-1 can change a matching flip (P5) or investigate half-turns and partner testing (P6-7). Grades 2-3 can choose reflection or half-turn constructions and edits (P4-6). Grades 4-5 can return to forced compositions (P3-4), or spend a visit finding all symmetries of one border (P5-6).

Entry key: K-1 Problem 1

A row of Border T copies spaced 4 cm apart matches after a 4 cm slide and fails after a 2 cm slide. Keep the original fixed and compare several repeats; explain how the pattern continues. Children may compare distances without measuring. The continuation key on page 4 covers Problems 2-7; the older keys keep their original numbers.